

Papilionem Complete Guidebook

Built from the current implemented milestone branch on 2026-04-12.

This guidebook is meant to do two jobs at once:

- teach an average player or tester how the game works
- preserve the exact internal logic, tuning, and system boundaries for future reference

Important note: when this guide talks about the game's neuro layer, it does *not* mean a machine-learning neural network. Papilionem uses a simulated life-model made of drives, emotions, memory, social edges, routines, interpretation, distortion, genetics, and lifecycle state.

Guidebook shape
page 1: table of contents
part 1: how to play and read the garden
part 2: how the living simulation works
part 3: internal systems, save/audit, battle
part 4: visual supplement + exact config tables + glossary

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1. Guide Overview And How To Use This Book

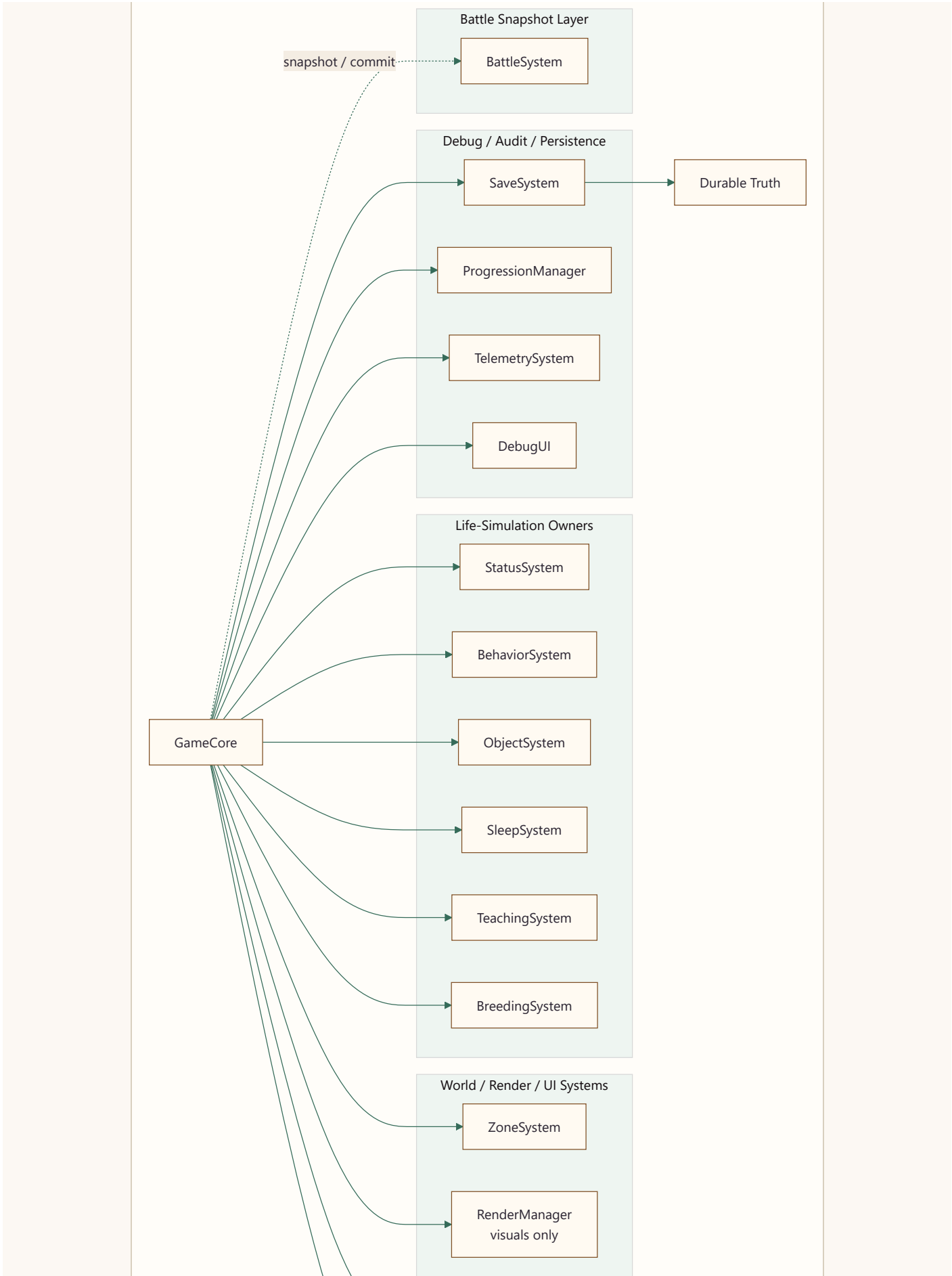
Who this is for

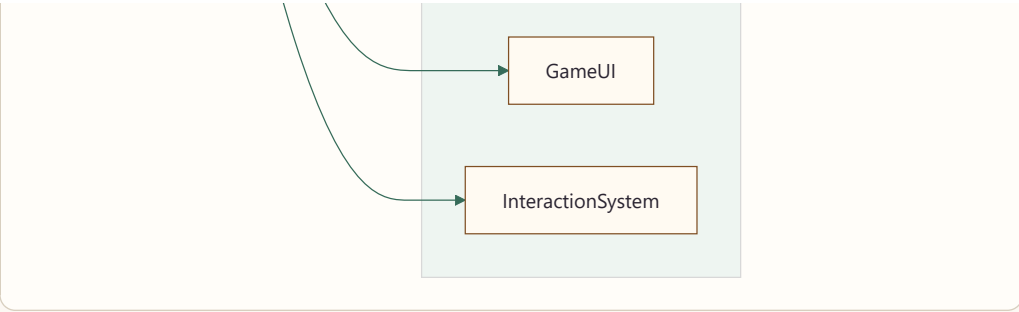
- players who want to understand what they are seeing
- testers who need to reproduce and describe behavior
- developers who need exact system logic and tunable values
- future-you when you want one place that explains the whole project

How it is organized

- the early sections are plain-language and player-facing
- the middle sections explain the life-simulation model
- the later sections cover persistence, audit tools, battle, and assets
- the appendix contains exact config values pulled from code

2. What The Game Is And How It Flows

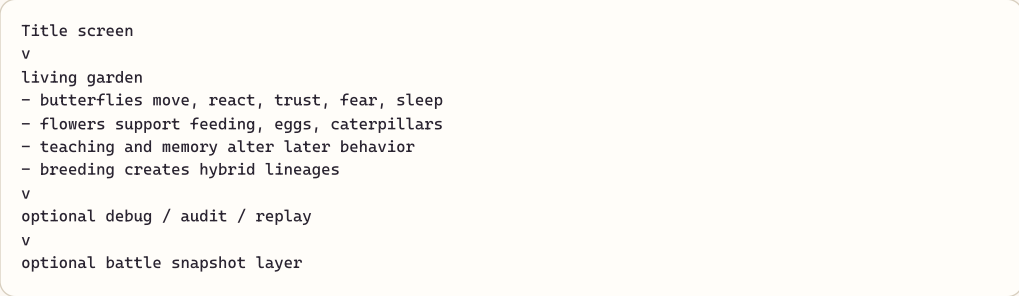




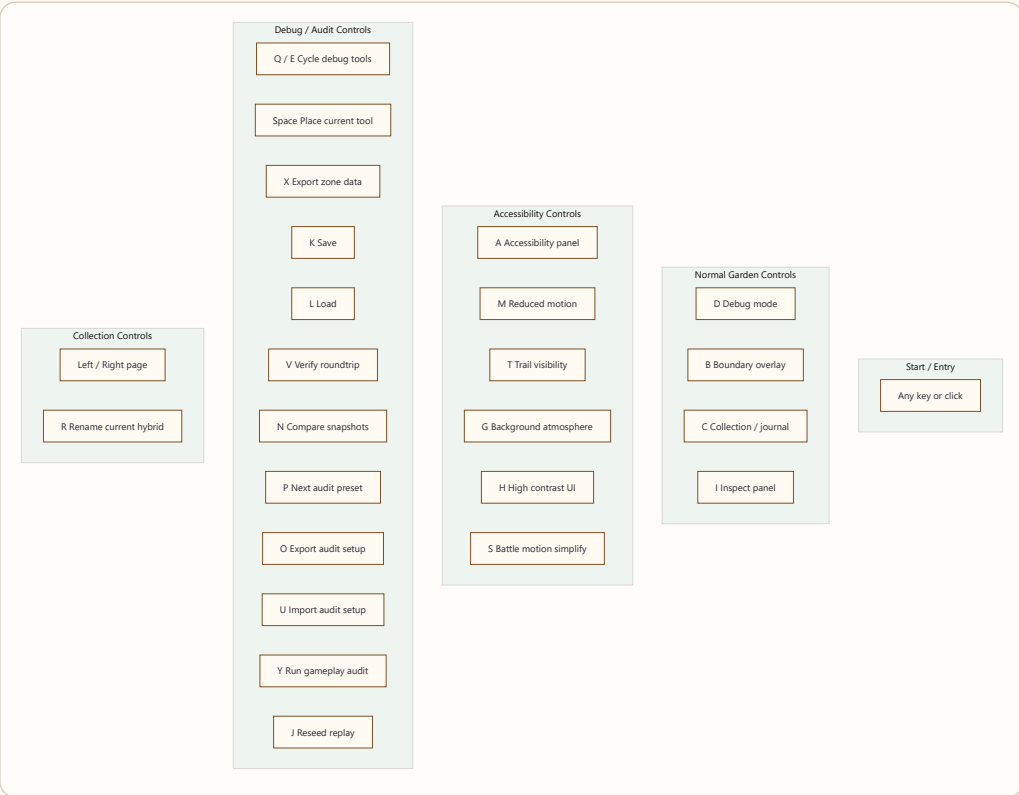
Papilionem is a living-garden simulation first. The butterflies are not just decorative sprites. Each one carries a persistent internal model that shapes how it moves, whom it trusts, how it learns, how it sleeps, and what it can pass on to offspring.

The short version: the player enters a garden, influences butterflies mainly through presence and care, and watches a long-running social, genetic, and lifecycle simulation unfold over time.

High-level flow



3. All Controls



Normal play controls

Normal controls

Setting	Value
Any key / click	Leave title screen and enter garden
D	Toggle debug mode
B	Show boundary overlay while held
C	Open butterfly collection / journal
I	Open inspect panel
A	Open accessibility panel
M	Toggle reduced motion
T	Cycle trail visibility
G	Cycle background atmosphere
H	Toggle high-contrast UI
S	Toggle battle motion simplify
Left / Right	Change collection page
R	Rename current hybrid in collection

Debug and audit controls

Debug controls

Setting	Value
Arrow keys	Move debug cursor on isometric grid
Q / E	Cycle debug tools
Space	Place current debug tool
X	Export zone data
K	Save game state
L	Load game state
V	Verify save/load roundtrip
N	Compare recent snapshots
P	Load next audit preset
O	Export audit setup
U	Import audit setup
Y	Run gameplay audit
J	Reseed replay session

4. World, Camera, Rendering, And Readability

The shipped build is a single-zone foundation, but the systems are already built as if the world can grow into multiple zones later. The camera and render layer know about view modes and readability rules.

Canvas: 800×450

Grid: 18×18

Tile: 18×9 isometric

Modes: overview, focused-garden, battle

- Focused garden mode allows more background motion and decorative presence.
- Overview mode suppresses moving background effects so the garden stays readable from afar.
- Battle mode removes moving background effects and afterimage trails so tactical readability wins over decoration.
- Accessibility toggles can further reduce motion, raise contrast, and simplify visual density.

Internal rule: rendering changes appearance only. It never changes gameplay range, collision, ownership, or state truth.

5. Entities And Lifecycle Objects

Butterflies

Butterflies are the main living agents. They hold personality traits, a life-simulation state, ability state, breeding state, social history, and long-term progression visibility.

Flowers

Flowers are consumable, non-carryable garden objects with their own lifecycle. They bloom, mature, wilt, and dissolve. They can also become lifecycle hosts for eggs and chrysalis stages.

Flower lifecycle defaults

Setting	Value
stageDurations.bloom	1200 frames (~20s at 60fps)
stageDurations.mature	2400 frames (~40s)
stageDurations.wilting	1200 frames (~20s)
stageDurations.dissolve	360 frames (~6s)
flower types	daisy, tulip, bush, lavender, sprout
resource tags	nectar, care, garden-object

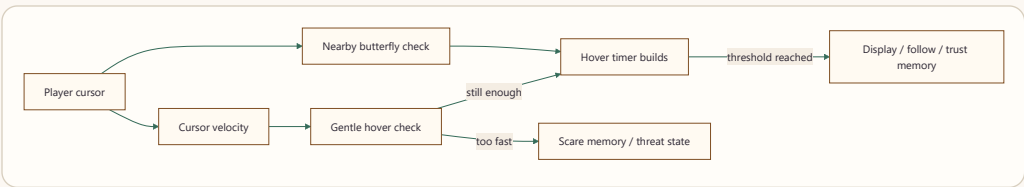
Caterpillars

Caterpillars are a real intermediate life stage, not just a note in the breeding system. They seek a food flower first, then seek a second flower to become a chrysalis host. If they fail to find what they need in time, they can starve and die.

Caterpillar behavior defaults

Setting	Value
initial phase	seekingFood
follow-up phase	seekingChrysalis
phase timeout	10800 frames (~3 minutes)
speed	0.35
lifecycle.stage	larval / phase-driven

6. What The Player Actually Changes



The player does not command butterflies directly like an RTS unit. The core player influence is the cursor and the conditions the player creates around the garden.

- A calm cursor can build gentle hover interactions and trust-related memory.
- A fast cursor can startle butterflies and create danger memory.
- Feeding and flower contact create object and outcome memories.
- Debug mode can directly create scenarios, but normal play mostly nudges the living systems rather than bypassing them.

7. Archetypes, Traits, Abilities, And Spawn Logic

Butterfly roster

Setting	Value
friendly	warm, trusting, support-oriented
cautious	delicate, patient, sparkle-oriented
energetic	fast, excitable, state-boosting
skittish	nervous, reactive, trust-cascade capable
wise	calm, teaching-oriented
mystic	sleep-comfort and calm aura oriented
golden	special legendary presence
hybrid	bred mixture with inherited traits and one parent ability

Shared trait axes

Setting	Value
speed	movement feel and pacing
jitteriness	movement noise and nervousness
trustPropensity	how likely trust is to build
trustSpeed	how quickly trust changes
scareThreshold	how easy it is to scare the butterfly
happinessBonus	how strongly positive outcomes raise mood

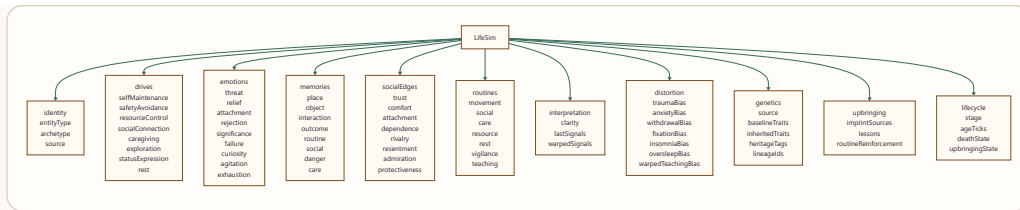
Ability mapping

Setting	Value
friendly	Warm Welcome: healing and panic support aura
cautious	Delicate Pink: sparkle / flower-oriented presence
energetic	Electric Violet: movement speed and energetic state boost
skittish	Nervous Jewel: trust cascade on care event
wise	Ancient Scholar: teaching pulse and lesson support
mystic	Twilight Dancer: sleep comfort and recovery support
golden	legendary special crown state
hybrid	one chosen parent ability inherited into hybrid genome

Current base spawn weights

Setting	Value
friendly	40
cautious	15
energetic	15
skittish	10
wise	10
mystic	10
golden	special, not part of normal weighted spawn

8. The Neuro-Social Life-Simulation Model



This is the heart of the game. Every major living agent is represented by a structured life-simulation state.

Plain-language summary: instead of a black-box AI brain, the game stores explicit pressures, feelings, memories, relationships, habits, inherited traits, and life stage. Those pieces are visible, inspectable, and persistent.

Drives

Drives are long-running pressures: self-maintenance, safety avoidance, resource control, social connection, caregiving, exploration, status expression, and rest.

Emotion channels

Emotion channels are current-state readings such as threat, relief, attachment, rejection, significance, failure, curiosity, agitation, and exhaustion.

Memory packets

Memories are concrete packets, not vague story text. Each packet has a family, subject, valence, strength, recency, reinforcement count, emotional tags, and metadata.

Social edges

Social edges are longer-term relationship summaries: trust, comfort, attachment, dependence, rivalry, resentment, admiration, and protectiveness.

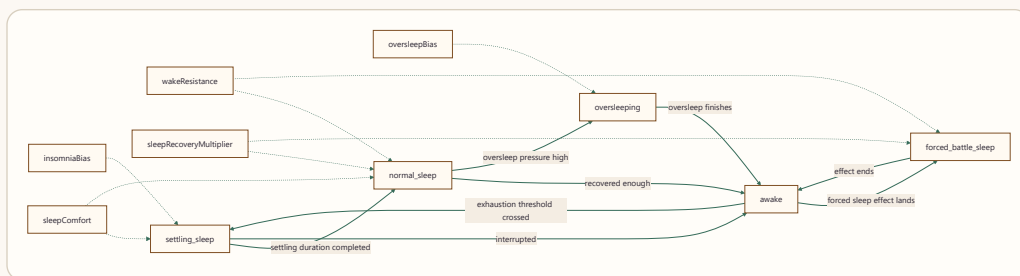
Routines

Routines are reinforced habits: movement, social, care, resource, rest, vigilance, and teaching.

Interpretation and distortion

Interpretation tracks clarity and signal handling. Distortion biases such as trauma, anxiety, insomnia, or warped teaching do not create separate disconnected systems. They bend the shared systems over time.

9. Sleep, Oversleep, Comfort, And Wake Rules



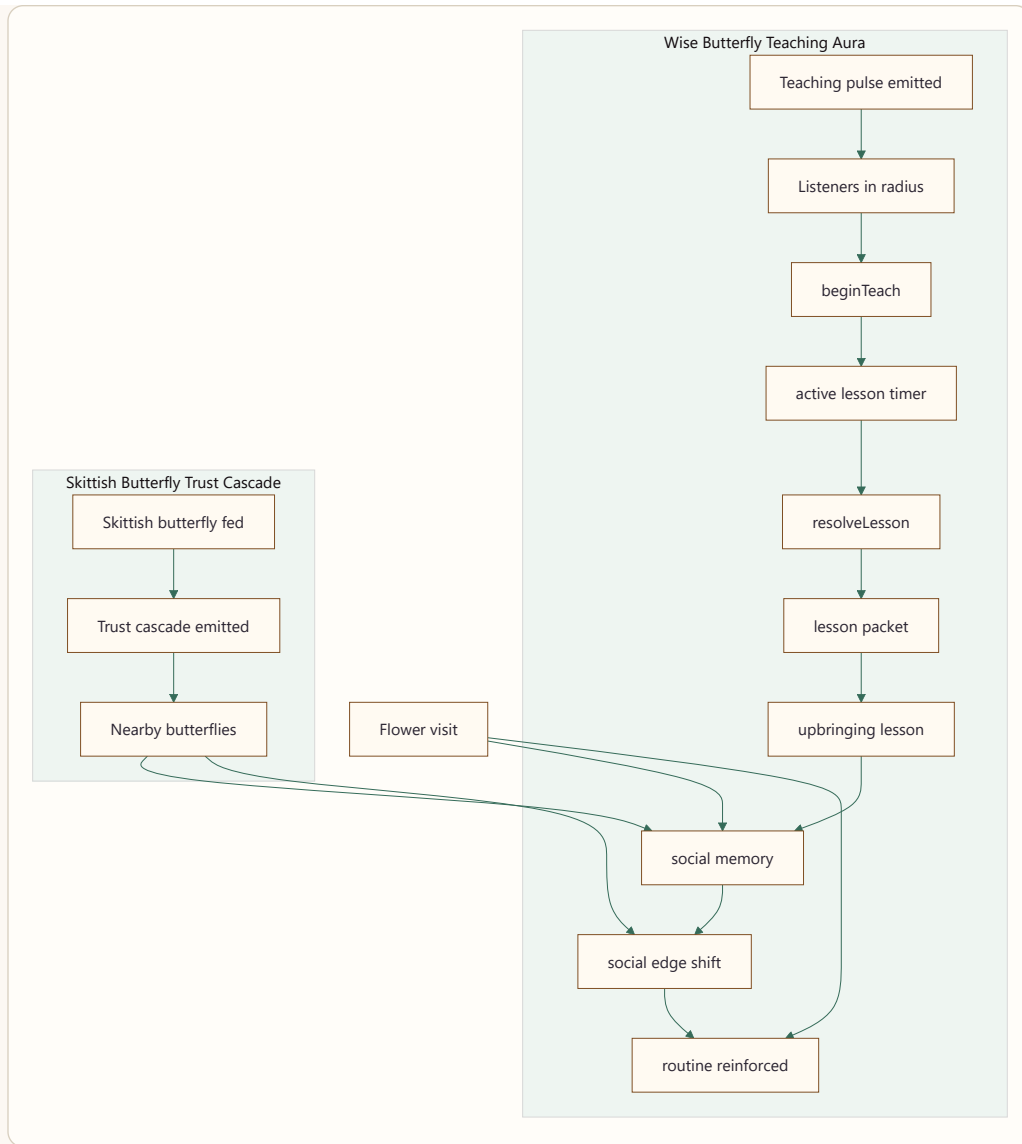
Sleep is a full system with owned state transitions. Butterflies do not just have a boolean asleep flag. They can be settling, normally asleep, oversleeping, or in forced battle sleep.

- Exhaustion grows passively while awake.
- Rest drive, insomnia bias, and comfort influence how easy it is to settle.
- Sleep assist sources and sleep-related status modifiers increase comfort or recovery.
- Oversleep is not random noise; it is driven by oversleep pressure and oversleep bias.
- Forced battle sleep is intentionally separated from normal garden sleep so battle effects do not accidentally use oversleep rules.

Stored sleep state fields

Setting	Value
subtype	current sleep subtype or null
exhaustion	current exhaustion value
sleepPressure	rest-drive-linked sleep pressure
sleepComfort	comfort gathered from assists and statuses
wakeDrive	wake pressure against staying asleep
oversleepPressure	pressure to remain asleep longer than needed
oversleepHabit	distortion-linked oversleep predisposition
assistSources	queued sleep assists for this update
settlingSeconds	time spent settling
asleepSeconds	time spent asleep
lastSleepStartSeconds	last sleep start timestamp
lastWakeSeconds	last wake timestamp
lastWakeReason	why the entity woke up

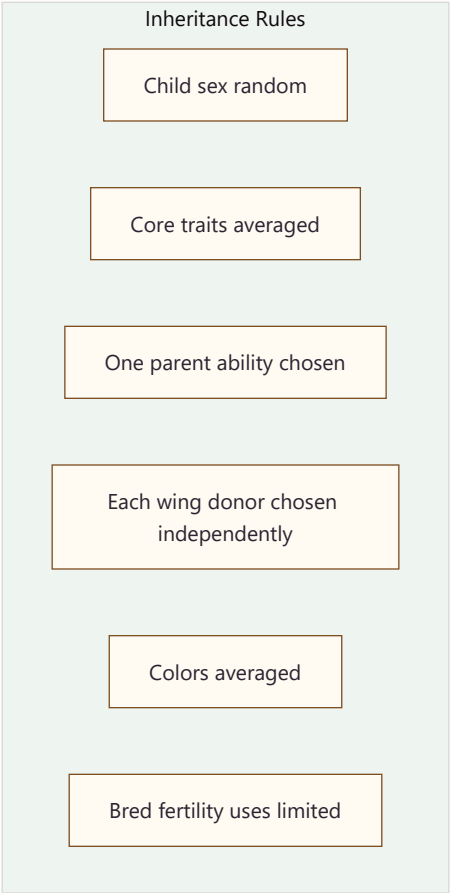
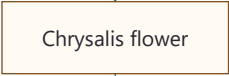
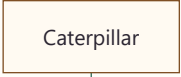
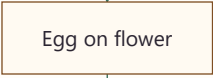
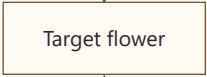
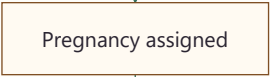
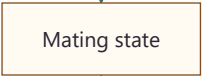
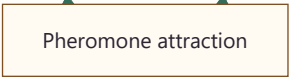
10. Teaching, Trust, Routines, And Social Pacing

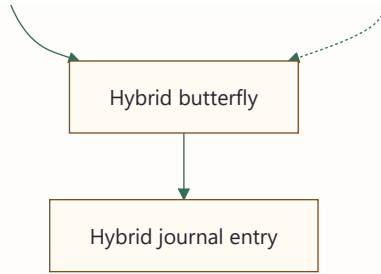


Teaching is a real continuity system. A wise butterfly's aura begins a lesson, the lesson runs for a tunable duration, then the result is turned into a packet, upbringing note, social memory, edge adjustment, and routine reinforcement. A skittish butterfly's trust cascade is also persistent. It does not just flash a temporary effect; it creates memory and relationship fallout in nearby butterflies.

- Flower visits can create object and outcome memory plus resource-routine reinforcement.
- Following the player cursor can create trusted-cursor interaction memory.
- Being scared by the player cursor can create danger memory tied to the cursor.
- Teaching can raise interpretation clarity and teaching-related routine strength.

11. Genetics, Breeding, Lineage, And Upbringing





Breeding is not a single spawn action. It is a staged lifecycle: attraction, mating, pregnancy, egg placement, caterpillar, chrysalis, adult hybrid, and journal entry.

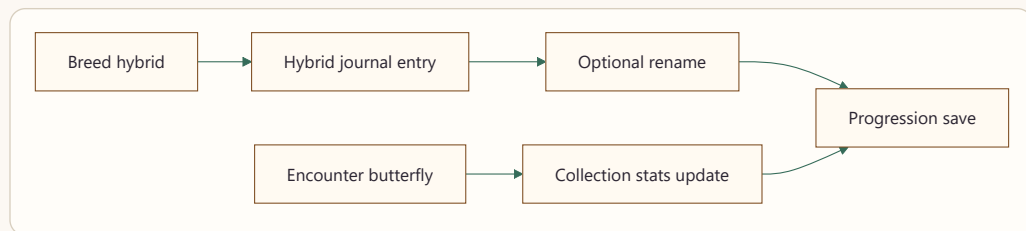
- Males and females must both be eligible, free, and within pheromone radius.
- Distance chooses the best attraction match among eligible pairs.
- Pregnancy is assigned to the female and targeted toward an appropriate flower.
- Egg and cocoon hatch timing is tunable and stored.
- Hybrid adults inherit averaged core traits, blended colors, one chosen parent ability, and independently chosen wing donors.

Genes vs upbringing: genes set the baseline predisposition. Upbringing does not erase that baseline; it reinforces, suppresses, clarifies, or distorts it over time.

Hybrid lifecycle artifacts

Setting	Value
pregnancy	stored on female butterfly until egg lay
lifecycleData	carries lineage and inheritance context through egg, caterpillar, and chrysalis stages
hybridGenome	stored on resulting hybrid butterfly
hybridJournal entry	stored in progression data and collection layer

12. Progression, Collection, And Hybrid Journaling



The progression layer turns the simulation into an archive. It stores encountered butterflies, collected butterflies, collection stats, hybrid journal entries, and the next hybrid id. The journal also supports renaming hybrids so discoveries feel personal and trackable.

Progression storage

Setting	Value
encounteredButterflies	set of seen types
collectedButterflies	set of collected types
butterflyCollectionStats	aggregate collection stats
hybridJournal	ordered hybrid entry list
nextHybridId	next journal id counter

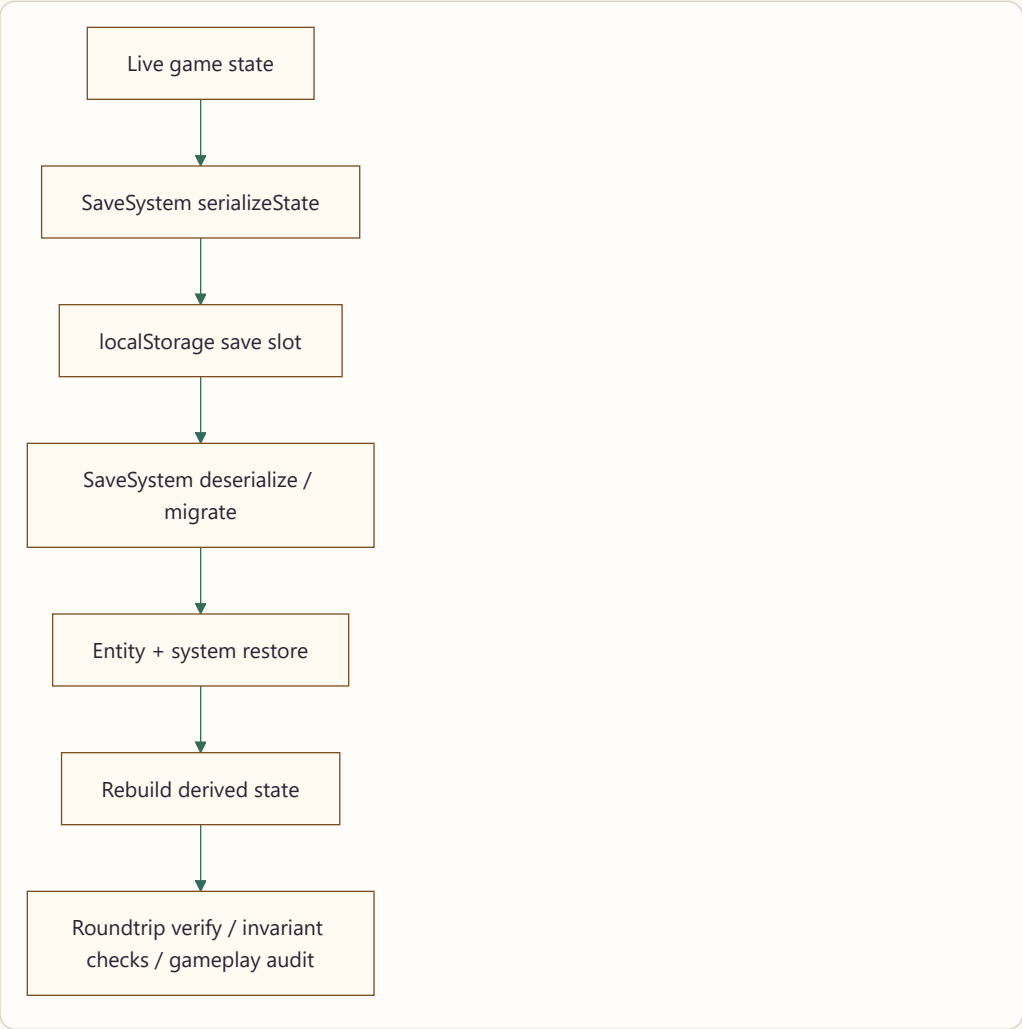
13. Internal Logic Ownership And Boundaries

One owner per truth is the main internal safety rule. This is how the game avoids double-applying logic or letting two systems disagree about the same state.

Owner map

Setting	Value
ZoneSystem	world zones, focus, and mode context
StatusSystem	timed effects, auras, cooldowns, charges, immunities, and aggregated modifiers
BehaviorSystem	current action runtime and target choice
ObjectSystem	carry, drop, delivery, and object ownership
SleepSystem	sleep transitions and sleep visual state inputs
TeachingSystem	lesson packets, trust cascades, social teaching fallout
BreedingSystem	mating, pregnancy, eggs, caterpillars, chrysalis, adult hybrid spawn
SaveSystem	durable serialization and migration
TelemetrySystem	update/render metrics
RenderManager	visuals only
BattleSystem	isolated battle snapshot layer

14. Save / Load, Replay, Audit, And Telemetry



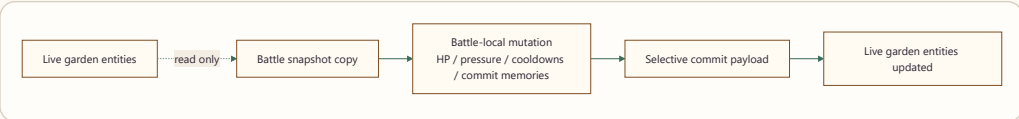
Persistence in Papilionem is designed around durable truth plus rebuilt derived state. Butterflies, flowers, caterpillars, progression, runtime values, and system-owned durable sub-states are serialized. Cheap derived values are reconstructed after load.

- SaveSystem captures butterfly, flower, and caterpillar state plus progression and selected runtime values.
- Sleep state is synchronized into life-sim lifecycle fields before save so the durable record stays honest.
- Roundtrip verification saves, reloads, rebuilds, and compares durable truth.
- Gameplay audit chains snapshots, invariant checks, and persistence checks into one pass.
- Replay metadata tracks session ids, seeds, and audit markers for reproducibility groundwork.

Telemetry

Telemetry records recent update and render samples, including total update time, foundation time, entity time, particle time, render time, particle render time, entity counts, time scale, and view mode. It is meant to support optimization without changing game truth.

15. Battle Snapshot Layer



Battle is intentionally isolated from the garden. When a battle starts, live entities are cloned into a battle snapshot. HP changes, pressure, retreat, cooldown spending, charge spending, and battle-generated social or memory outcomes are all recorded inside that snapshot first. Only selected results are committed back into the live garden state. This keeps battle experimentation from corrupting normal simulation truth.

16. Sprite And Asset Pipeline

Runtime rendering uses the repository's asset folders, while the sprite manager slices, cleans, and composes those assets into renderable butterflies and lifecycle visuals.

Sprite pipeline

Setting	Value
butterfly wing base path	assets/butterflies/
wing file organization	separate male/female full wing composites per archetype
body sprite	ephemera-butterfly-body-.png
antenna sprite	ephemera-butterfly-antenna.png
cocoon sprites	assets/cocoons/cocoon-unhatched.png and cocoon-hatched.png
caterpillar frames	assets/caterpillars/caterpillar-crawl1/2/3.png
wing split point	x=900, y=600
background cleanup rule	near-black pixels converted to transparent

The render spec for hybrids can mix wing donors per wing, so hybrid visuals are not limited to a single parent's full wing set.

17. Visual Supplement: Sprite Anatomy, Archetypes, Hybrids, And Lifecycle

This section uses the actual runtime sprite assets from the project to show how the butterfly visuals are built, how the male and female versions differ at render time, how hybrids mix donor wings, and how the caterpillar-to-cocoon-to-adult lifecycle is displayed.

- The anatomy plate shows the body sprite, antenna sprite, wing sheet split points, and anchor facts used by the sprite manager.
- The comparison plates show assembled in-game male and female renders for every archetype, not just raw wing sheets.
- The hybrid plate uses mixed donor wings so the visual inheritance logic can be read at a glance.
- The lifecycle plate shows caterpillar frames, cocoon states, adult composition order, and wing-spread-based adult motion.

Visual Supplement 01 · Sprite Anatomy

The butterfly is built from separate runtime layers: body, antenna, and a full wing sheet split into four independent pieces.

Body + antenna source sprites

These are separate images. The body sets the center. The antenna image is anchored to the body, not baked into the wing sheet.



body sprite
1080×1080



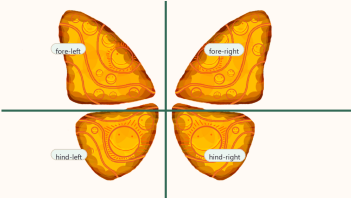
antenna sprite
1080×1080

Body center
540, 540

Antenna mounts
left 488,290 right 559,290
The antenna sprite is anchored to these body points.

Wing sheet split and anchor logic

The runtime sheet is cut at x=900 and y=600, producing fore-left, fore-right, hind-left, and hind-right pieces that can come from different donors in hybrids.

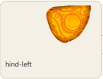




fore-left



fore-right



hind-left



hind-right

Composition facts

These labels are taken from the live sprite manager and show what the renderer actually uses.

Wing sheet size: 1920×1080

Body sprite size: 1080×1080

Antenna sprite size: 1080×1080

Split points: x=900, y=600

Render order: body → antenna → hindwings → forewings

Hybrid children can pull each wing piece from a different parent donor

Wing anchor points on body

foreLeft: body 375,549 piece 841,500

foreRight: body 700,549 piece 68,500

hindLeft: body 375,682 piece 852,-18

hindRight: body 700,682 piece 51,-18

Sprite anatomy plate. Shows the separate body and antenna assets, a representative wing sheet split at x=900 and y=600, and the runtime anchor facts that let the renderer assemble the butterfly.

Visual Supplement 02 · Archetype Male/Female Comparisons

Each row uses the real in-game sprite assets and the same sex-based scale rules the renderer uses: males use a smaller body scale, while females use a larger wing scale.

Male render

Female render

Warm Welcome
friendly



Delicate Pink
cautious



Electric Violet
energetic



Nervous Jewel
skittish



Archetype male/female comparison plate, core roster. Shows the assembled male and female versions of Warm Welcome, Delicate Pink, Electric Violet, and Nervous Jewel using the same sex-based sprite scaling the game uses.

Visual Supplement 03 · Archetype Male/Female Comparisons

Each row uses the real in-game sprite assets and the same sex-based scale rules the renderer uses: males use a smaller body scale, while females use a larger wing scale.

Male render

Female render

Ancient Scholar

wise



Twilight Dancer

mystic



Legendary One

golden



Archetype male/female comparison plate, special roster. Shows the assembled male and female versions of Ancient Scholar, Twilight Dancer, and Legendary One, preserving the visual distinction between ordinary archetypes and the special golden form.

Visual Supplement 04 · Hybrid Inheritance Visual

Hybrid butterflies do not inherit a single parent sprite sheet. Each wing can choose its donor independently, while traits and colors are blended and one parent ability is chosen.

Mother donor

Friendly female / Warm Welcome



Hybrid child

Example child uses female body/wing scaling, averaged traits, averaged colors, and mixed wing donors.



Father donor

Energetic male / Electric Violet



Per-wing donor map

Each wing key chooses mother or father independently.

fore-left	mother - friendly female		
fore-right	father - energetic male		
hind-left	father - energetic male		
hind-right	mother - friendly female		
			
FL	FR	HL	HR

What else inherits

The visual donor map is only one part of breeding. The lifecycle system also chooses traits, colors, sex, and ability.

Runtime inheritance summary

- Child sex is chosen randomly and determines body/wing scale profile.
- Core traits are averaged across both parents.
- The visible body colors are averaged pairwise.
- One parent ability is chosen for the child.
- The hybrid genome stores wingDonors per wing key.

Hybrid inheritance plate. Shows one worked hybrid example with a friendly female donor, an energetic male donor, a mixed child, and a per-wing donor map so the visual inheritance rule is explicit.

Visual Supplement 05 · Animation And Lifecycle Breakdown

The lifecycle visuals are assembled from multiple asset types: caterpillar frames, cocoon states, and a layered adult butterfly whose wings animate by spread instead of swapping whole body sprites.

Caterpillar crawl frames

The larval stage uses a small frame cycle rather than the adult wing-spread system.



Cocoon states

The chrysalis stage has explicit unhatched and hatched sprites.



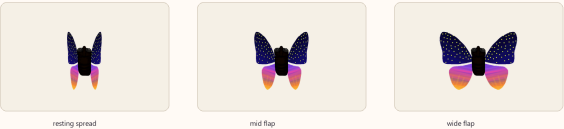
Adult composition order

The runtime render draws body first, then antenna, then hindwings, then forewings. Forewings sit on top.



Adult wing motion

Adult butterfly animation comes from changing wing spread and tilt, not from swapping a separate frame sheet.



Animation and lifecycle plate. Shows caterpillar crawl frames, cocoon states, adult composition order, and the adult wing-spread animation approach that creates flap motion without swapping a separate butterfly frame sheet.

18. Exact Tunable Config Tables

These tables are generated from the live `core/config.js` structure so the guidebook reflects the implemented values instead of a hand-maintained copy.

Rendering

Setting	Value
useSprites	true
butterflyVisualScale	1.6

Canvas

Setting	Value
baseWidth	800
baseHeight	450
targetWidth	800
targetHeight	450
backgroundColor	#f4e8dc

Grid

Setting	Value
cellSize	16
gridWidth	18
gridHeight	18
debugGridSize	32
gridOffset.x	10
gridOffset.y	8

Isometric

Setting	Value
tileWidth	18
tileHeight	9
offsetX	400
offsetY	100
bounds.minX	0
bounds.maxX	17
bounds.minY	0
bounds.maxY	17

Entities: limits and defaults

Setting	Value
maxButterflies	12
maxFlowers	6
heightOffset.butterfly	10
heightOffset.flower	0
heightOffset.pixel	0
butterfly.size	12
butterfly.speed	0.008
butterfly.wanderTimer.min	40
butterfly.wanderTimer.max	90
butterfly.maxWanderDistance	3
butterfly.lifetime	10000
butterfly.personalities	brave, cautious, curious
butterfly.colors	255, 140, 60, 255, 220, 120, 220, 100, 150, 255, 180, 200, 150, 120, 200, 200, 170, 255, 100, 180
flower.types	daisy, tulip, rose, sunflower, lily
flower.stemHeight	16
flower.stageDurations.bloom	1200
flower.stageDurations.mature	2400
flower.stageDurations.wilting	1200
flower.stageDurations.dissolve	360
flower.palettes	{ "petals": [255, 180, 120], "center": [255, 240, 180] }, { "petals": [255, 150, 200], "center": [255, 255, 220] }, { "petals": [200, 150, 255], "center": [255, 230, 150] }, { "petals": [255, 220, 150], "center": [255, 255, 200] }, { "petals": [255, 255, 240] }

Particles

Setting
maxParticles
gravity
pixelSize
bounce
friction
types.scale.lifetime
types.scale.fadeSpeed
types.joy.lifetime
types.joy.fadeSpeed
types.stress.lifetime
types.stress.fadeSpeed
types.happy.lifetime
types.happy.fadeSpeed
types.happy_visual.lifetime
types.happy_visual.fadeSpeed

Color pools

Setting	Value
radius	20
requiredPixels	50
pulseSpeed	0.1
spawnDelay	180
gridSize	40

Interaction

Setting
stillFramesRequired
cursorZoneRadius
butterflyZones.brave.comfort
butterflyZones.brave.flee
butterflyZones.cautious.comfort
butterflyZones.cautious.flee
butterflyZones.curious.comfort
butterflyZones.curious.flee

Effects

Setting	Value
shadowOpacity	50
glowPulseSpeed	0.1
swaySpeed.min	0.02
swaySpeed.max	0.03
swayAmount	0.1

Debug

Setting
enabled
showGrid
showZones
showCoordinates
showFPS
gardenTimeControls.enabledInNormalPlay
gardenTimeControls.debugSpeedPresets
auditTools.scenarioPresets
auditTools.snapshotDiff
auditTools.invariantChecker
auditTools.eventTimeline
auditTools.saveLoadVerifier
auditTools.screenshotNotes

Accessibility

Setting	Value
reducedMotion	false
battleMotionSimplify	true
highContrastUI	false
colorblindSafeIndicators	true
strongSelectionOutlines	true
trailVisibility	full
backgroundAtmosphere	full
statusIndicatorDensity	simplified
uiScale	1

Simulation

Setting	Value
defaultTimeScale	1
battleTimeScales	1, 2
frameRate	60
fixedDeltaSeconds	0.016666666666666666

Balance: sleep

Setting
assistStrengthDefault
movementMultiplierSettling
wingAnimationMultiplierSettling
wingAnimationMultiplierAsleep
visualYOffsetSettling
visualYOffsetAsleep
visualTiltSettling
visualTiltAsleep
passiveExhaustionBaseGain
passiveExhaustionRestMultiplier
passiveExhaustionInsomniaMultiplier
settleThresholdBase
settleThresholdFloor
settleThresholdInsomniaMultiplier
settleThresholdComfortMultiplier
settlingDurationSeconds
normalRecoveryBaseRate
normalRecoveryComfortMultiplier
oversleepPressureGainMultiplier
wakeThresholdBase
wakeThresholdFloor
wakeThresholdResistanceMultiplier
wakeMinimumSleepSeconds
oversleepThresholdBase
oversleepThresholdBiasMultiplier
oversleepRecoveryRate
oversleepPressureDecayRate
forcedSleepRecoveryRate
passiveOversleepPressureDecayRate

Balance: social

Setting
teachingLessonDurationSeconds
teachingPulseRadius
teachingPulseMemoryValence
teachingPulseMemoryStrength
teachingPulseEdgeTrust
teachingPulseEdgeAdmiration
teachingPulseEdgeComfort
teachingPulseRoutineReinforcement
trustCascadeRadius
trustCascadeMemoryValence
trustCascadeMemoryStrength
trustCascadeEdgeTrust
trustCascadeEdgeComfort
trustCascadeEdgeAdmiration
lessonUpbringingStrength
lessonRoutineReinforcement
lessonInterpretationClarityGain
lessonMemoryValence
lessonMemoryStrength
lessonEdgeTrust
lessonEdgeAdmiration
lessonEdgeComfort
listenerTeachingRoutineReinforcement
teacherTeachingRoutineReinforcement
teachingBoostFrames

Balance: hybrid

Setting	Value
pheromoneRadius	13
matingDistance	16
matingDurationFrames	15
pheromoneCooldownFrames	21
adultHardCap	50
eggHatchFrames.min	28
eggHatchFrames.max	39
cocoonHatchFrames.min	28
cocoonHatchFrames.max	39
bredFertilityUses	1

World

Setting	Value
layout	single-zone-foundation
overviewMode	false
viewModes	overview, focused-garden, battle
zones	{ "id": "garden-core", "label": "Garden Core", "kind": "garden", "poolAllowed": true, "doorwayIds": [], "adjacentZoneIds": [], { "minX": 0, "maxX": 17, "minY": 0, "maxY": 17}, "renderProfile": { "movingBackgroundEffectsInFocus": true, "movingBackgroundEffectsInOverview": false, "afterimageTrailsInFocus": true, "a

Registries

Setting	Value
actionFamilies	idle, wander, seek_resource, care_for_vulnerable, teach_or_listen, sleep, socialize, reposition
statusFamilies	healing_received_bonus, panic_resistance, aggression_suppression, attack_speed_bonus, movement_energetic_state_boost, reposition_guidance, cooldown_intelligence, sleep_comfort_bonus, wake_reforced_sleep_immunity, sleep_recovery_multiplier
sleepSubtypes	settling_sleep, normal_sleep, oversleeping, forced_battle_sleep
battleStates	inactive, snapshotting, active, resolving, committing
persistenceFieldClasses.durable	durable
persistenceFieldClasses.derived	derived
persistenceFieldClasses.runtime	runtime

System order

Setting	Value
phaseFoundationOrder	zoneSystem, statusSystem, behaviorSystem, objectSystem, sleepSystem, teachingSystem, battleSystem, sa

19. Glossary

Glossary

Setting	Value
Archetype	A base butterfly type such as friendly, wise, or mystic.
Ability	A special effect identity tied to an archetype or inherited by a hybrid.
Drive	A long-running pressure such as rest, exploration, or safety.
Emotion channel	A current-state feeling value such as threat, curiosity, or exhaustion.
Memory packet	A stored event-like record with valence, strength, and tags.
Social edge	A durable relationship summary like trust or admiration.
Routine	A reinforced tendency to behave in a certain way.
Distortion	A bias that bends shared systems, such as anxiety or insomnia bias.
LifeSim	The full internal life-simulation state carried by an entity.
Hybrid genome	The inherited trait and visual source data for a bred hybrid butterfly.
Roundtrip verify	A save/load comparison that checks whether durable truth survives correctly.
Gameplay audit	A chained test pass that checks snapshots, invariants, save/load, and visible state.
Battle snapshot	A battle-local copy of live entities used so battle can mutate safely before commit.